WIRELESS EV CHARGER - SS-COMPENSATED 85 kHz WPT WITH TWO-STAGE AUTOMATIC FREQUENCY SEARCH

Power topology follows the Simulink reference (230 V AC, diode bridge, L-C DC link, 4-MOSFET full bridge, series C + coupled TX/RX coils + series C, diode bridge, C + R, battery).

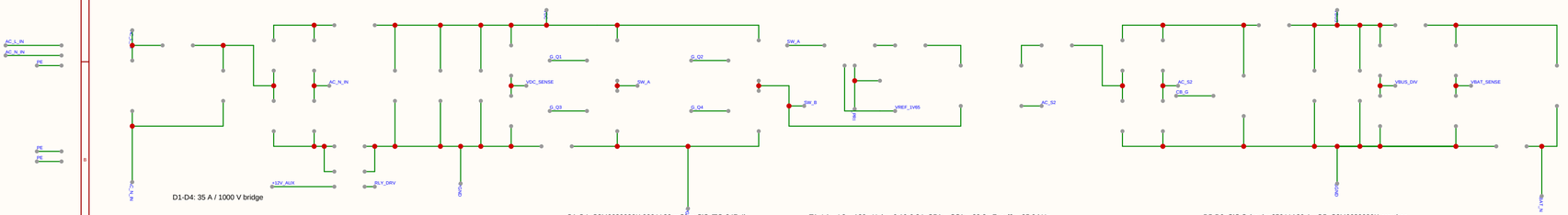
1 AC INPUT

2 RECTIFIER + PRE-CHARGE + L-C DC LINK

3 FULL-BRIDGE INVERTER

4 SERIES-SERIES RESONANT LINK

5 SECONDARY RECTIFIER + OVP CROWBAR + BATTERY (VEHICLE)



D1-D4: 35 A / 1000 V bridge

Q1-Q4: C3M0030090K 900 V 30 mOhm SiC (TO-247-4)

T1: L1 = L2 = 120 uH, k = 0.10-0.34; CP1 = CS1 = 29.2 nF -> f0 = 85.0 kHz

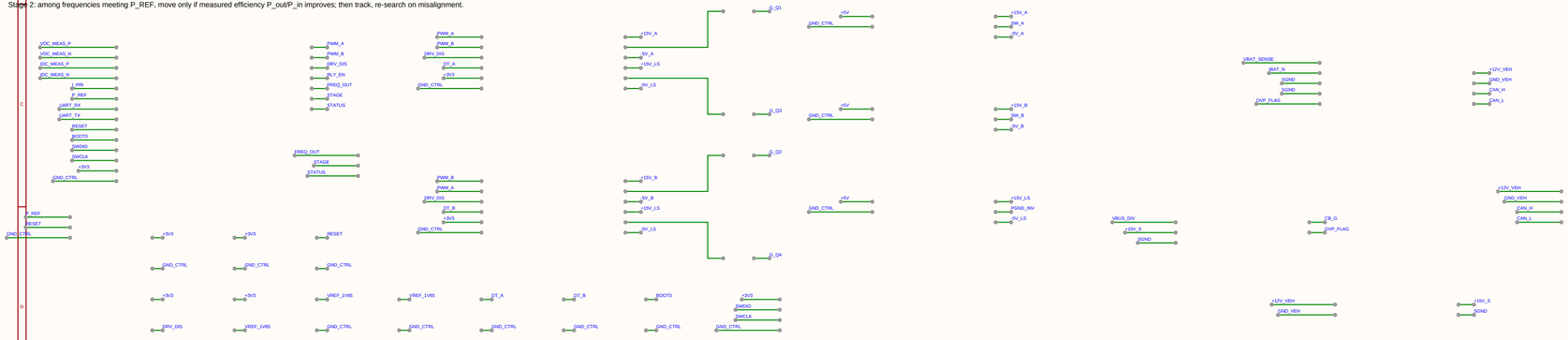
D5-D9: SiC Schottky 650 V / 20 A Q5: C3M0030090K crowbar

Ratings: F1 20 A T 250 VAC; RV1 275 VAC; NTC1 10 ohm, K1 30 A (bypass after pre-charge); L1 10 mH 25 A; C1 2200 uF 450 V bank (>=12 A rms ripple); C2 900 V film; R1 47k 5 W; R2 = 3 x 1 Mohm HV chain; RSH1/RSH2 3 W; CP1/CS1 29.2 nF PP film banks >= 2.5 kV rms / 40 A rms; C3 20 uF 600 V film; F2 20 A 500 VDC; RBUR 0.5 W; CT1 1:100.

6 CONTROL - Adaptive_Freq_Controller (STM32G474, HRTIM)

Stage 1: sweep 79-90 kHz and bisect until P_out = P_REF +-3 % within I_pri / I_bat limits (reports UNMET if impossible).
Stage 2: among frequencies meeting P_REF, move only if measured efficiency P_out/P_in improves; then track, re-search on misalignment.

7 ISOLATED GATE DRIVE - diagonal pairs: Q1 and Q4 on PWM_A, Q2 and Q3 on PWM_B



8 ISOLATED SENSING + AUX SUPPLY + PRE-CHARGE RELAY DRIVER

Values are engineering assumptions (not recovered from the original model): f0 = 85 kHz (79-90 kHz), L1 = L2 = 120 uH, C = 29.2 nF, P_REF = 3.3 kW, battery 300-380 V, I_pri <= 35 A, I_bat <= 12 A.
LIMITATION: with an uncontrolled diode front end (V_DC about 300 V) SS frequency control alone reaches 3.3 kW inside the current limits only at high coupling (k ~ 0.3-0.4); low-k operation needs a regulated DC link, phase-shift control or a coil redesign.
SAFETY: primary side is mains-referenced, resonant capacitors reach ~2.2 kV rms. Not included / not verified: EMI filter, PFC (IEC 61000-3-2), RCD/earth-leakage, insulation coordination, thermal design, hardware validation. Not reviewed for mains connection.
Protection: primary OCP = I_PRI -> STM32 COMP1 -> HRTIM fault (firmware); secondary OVP = U11 latch fires Q5 crowbar (SS: secondary short collapses primary current) + OVP_FLAG; F2 fuse; charging CC/CV only via P_REF from vehicle BMS (CAN -> Wi-Fi).

9 VEHICLE SIDE - telemetry / BMS gateway (P_out = V_bat x I_bat over Wi-Fi), OVP crowbar

Reference: user Simulink screenshots

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File : Wireless_EV_Charging

Title : Wireless EV Charger - Two-Stage Automatic Frequency Search

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